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Review

Burden of vaccine preventable diseases at large events

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ABSTRACT

Background: Large events or mass gatherings (MGs) are known to amplify the risk of infectious diseases, many of which can be prevented by vaccination. In this review we have evaluated the burden of vaccine preventable diseases (VPDs) in MGs.
Methods: Major databases like PubMed and Embase, Google Scholar and pertinent websites were searched by using MeSH terms and text words; this was supplemented by hand searching. Following data abstraction, the pooled estimate of the burden of VPDs was calculated when possible; otherwise a narrative synthesis was conducted.
Results: In the past, at religious MGs like Hajj and Kumbh Mela, cholera caused explosive outbreaks; but currently respiratory infections, notably influenza, are the commonest diseases not only at Hajj but also at World Youth Day and Winter Olympiad. The recent cumulative attack rate of influenza at Hajj is 8.7% (range 0.7–15.8%), and the cumulative prevalence is 3.6% (range: 0.3–38%). Small outbreaks of measles (13–42 cases per event) have been reported at sport, entertainment and religious events. A sizeable outbreak (>200 cases) was reported following a special Easter Festival in Austria. An outbreak of hepatitis A occurred following the 'Jam bands' music festival. Other VPDs including pneumococcal disease, pertussis and tuberculosis have been reported in relation to MG attendance.
Conclusion: VPDs not only affect the participants of MGs but also their contacts; vaccine uptake is variable and vaccine implementation is likely to have beneficial effects. Research to address the knowledge gaps surrounding VPDs at MGs is needed.

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1. Introduction

Q2 According to the World Health Organization (WHO) an organised or planned event can be classified as a mass gathering (MG) "if the number of people attending is sufficient to strain the planning and response resources of the community, state or nation hosting the event" [1]. MGs are held for various purposes including religious, political, socio-cultural, athletic and entertainment pursuits. Obviously, health and safety of the attendees at such an event are essential elements that concern both the host country and international public health agencies.

MGs are known to increase the risk of infectious diseases with the potential for a pandemic i.e., 'globalisation' of a pathogen following the event [2]. The risk and pattern of diseases at MGs are influenced by the features of the event, particular activities, and also characteristics of the participants including their immunity to infectious agents [3]. Excessive crowding, shared accommodation and prolonged exposure resulting from close contact with infectious individuals all increase the risk of transmission of infectious agents [4,5].

Intercontinental outbreaks of meningococcal disease associated with the Hajj pilgrimage in 1987 and 2000/2001, and more recently, rapid international spread of influenza and other respiratory pathogens are all well-known examples of health hazards

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at MGs [2,6]. Other notable examples are, outbreaks of influenza at a World Youth Day (WYD) [7], a Winter Olympiad [8] and a Rock Festival [9], and historically of cholera at Kumbh Mela, one of the largest gatherings in the world that involves ritual bathing in the Ganges in India [10]. These and other health hazards associated with MGs have been well reviewed [11–15]. It is likely that a substantial proportion of these infections can be prevented by immunisations. Vaccinations against meningococcal disease, influenza, yellow fever and polio are recommended for attendees at certain MGs [3,16]. However, the burden of vaccine preventable diseases (VPDs) associated with MGs has not been quantified through a formal review.

The aim of this review is to evaluate the burden of VPDs during MGs.

2. Search strategy

2.1. Search strategy and selection criteria

Studies were identified through searching electronic databases including; PubMed and Embase from database inception to April 30, 2015. We used a combination of MeSH terms and text words including: 'mass gathering' OR 'holiday' OR 'sport' OR 'festival' OR 'Hajj' OR 'FIFA', OR 'pilgrimage' AND 'infection' OR 'disease outbreak' OR' respiratory disease' OR 'influenza' OR 'communicable disease' AND 'vaccine' OR 'vaccination' OR 'immunisation' OR 'vaccine uptake'. Google Scholar and pertinent websites including the online Saudi Epidemiology Bulletin were also searched for relevant studies. Finally, manual searches were performed reviewing reference lists of included studies to identify additional potentially relevant papers.

Inclusion criteria were all observational studies that reported the rate of infectious diseases preventable by vaccine e.g., influenza, meningococcal and pneumococcal diseases, pertussis, diphtheria, measles, mumps, rubella, hepatitis A and B, tetanus, diphtheria and varicella. The target population of this review was attendees at MGs of any age, gender and country. Only English language studies were included.

We excluded case reports, case series and studies involving healthcare workers or that focussed on health services or infrastructure during MGs. Studies reporting infections not preventable by vaccine were also excluded.

A selective strategy was used for older papers that reported occurrence of repeated outbreaks such as cholera at Hajj and Kubh Mela whereby only the most comprehensive articles were included. Similarly, for the meningococcal outbreaks at Hajj, only the key outbreak reports were included.

Two authors (ASA and MA) independently reviewed the titles and abstracts to identify potentially relevant papers. Full texts of all potentially relevant papers were reviewed by both authors to determine those which met the selection criteria. The 'preferred reporting items for systematic reviews and meta-analyses (PRISMA)' statement was used to guide and report the search methodology (Fig. 1).

2.2. Data extraction

Two authors (ASA and MA) independently extracted the data from each study into a data extraction sheet which was divided in sections for each MG event, while a third author (HR) arbitrated when a discrepancy occurred. The following data were abstracted in each extraction sheet: study design, year and country of conducting the study, sample size, age group, diagnostic method used, type and prevalence of infection, vaccination rate and effectiveness of vaccine, if available. Whenever possible we provided a pooled estimate of the burden of VPDs otherwise a narrative synthesis was conducted.

3. Disease burden at MGs

3.1. Summary of included studies

Of 3036 records identified, 66 full texts were reviewed and 39 studies were finally included in this review (Fig. 1); of these, 31 studies reported the burden of VPDs at religious MGs (Hajj, Kumbh Mela, WYD, Taizé meeting and Easter Festival), 4 reported on sport events (Winter Olympics, Asian Youth Games, International Youth Sport Event and International Special Olympic Games) and the remaining 4 studies reported on entertainment MGs (music concert, International Dog Show and fun gatherings at Disney parks). The characteristics of these MGs have been listed in Table 1.

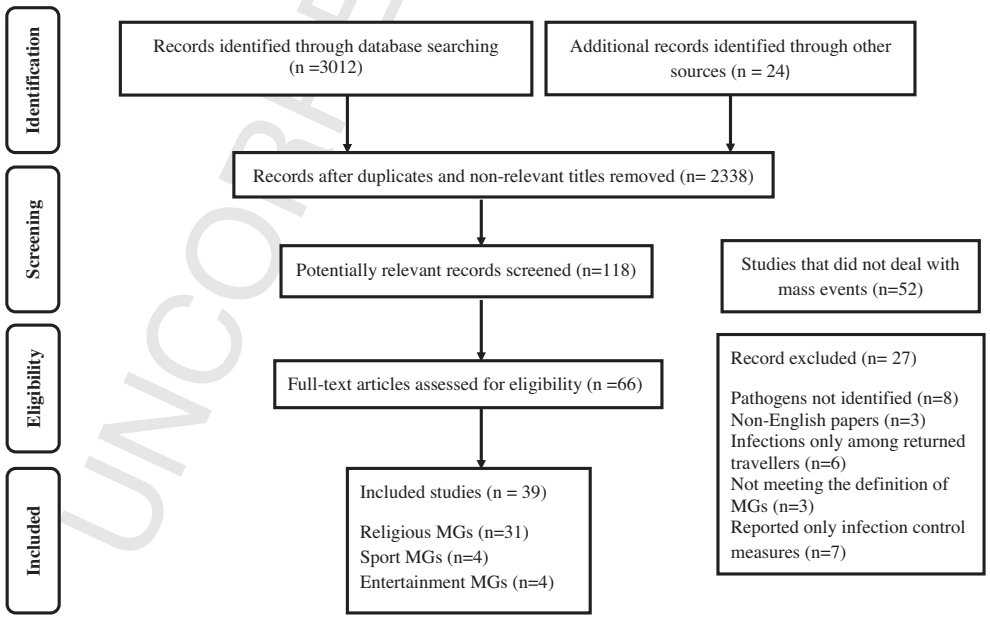

Fig. 1. Flow diagram of searching strategy.

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Table 1
Characteristics of the included mass gatherings.

MGs	Event type	Host country	Frequency	Event duration	Number of attendees	Major hazards
Hajj	Religious	Saudi Arabia	Annually	Peak of the event six days	2–3 million	Extreme crowding Air pollution Risk of blood-borne diseases Stampede and fire incidents
WYD	Religious	Variable	Every three years	Around 12 days	1–3 million	Crowding Risk of food-borne illnesses
Kumbh Mela	Religious	India	Every three years	55 days	100 million	Food and water sanitation issues Stampede incidents
Austrian Easter Festival	Musical	Southern Austria	Annually	One week	Thousands	Crowding Risk of air-borne infections
Taizé Meeting	Religious	Variable, centrally in Taizé, France	Annually, round the year	Days to weeks	Up to 5000	Risk of air-borne infections
Winter Olympic Games	Sports	Variable	Every four years	Around a month	About several millions	Risk of respiratory infections Coincidence with epidemics of seasonal influenza
International Youth Sporting Event	Sports	Variable	Irregular	About 10 days	265,000	Risk of air-borne, food and water-borne infections
Asian Youth Games	Sport	Variable	Every four years	Around a month	1000–2000	Risk of air-borne infections
International Special Olympics Games	Sport	Variable	Every 2 years	One week	40,000–150,000	Risk of air-borne, food and water-borne infections
Rock Werchter Festival	Musical	Belgium	Annually	4 days	6900–80,000	Risk of air-borne infections
Disney Parks	Entertainment	USA	Year around	Year around	About several millions	Risk of air-borne infections
Music concert "Jam bands"	Musical	USA	Annually	One week	1200–82,000	Food and water-borne infections Alcohol and drug use
International Dog Show	Entertainment	Variable	Annually	6 days	10,000	Risk of air-borne infections Risk of zoonotic diseases

WYD, World Youth Day.

Included studies reported outbreaks/diseases among participants of MGs that took place in Saudi Arabia, India, USA, UK, Qatar, Australia, Singapore, Slovenia and Belgium, between 1893 and 2014. Participants (i.e., travellers) from a number of countries were represented including, Saudi Arabia, India, France, Iran, Singapore, UK, USA, Australia, Slovenia, Belgium, Ghana, and elsewhere. Most reported diseases among the studies were air borne infections including influenza, measles, meningococcal disease and tuberculosis.

3.2. Religious MGs

3.2.1. Hajj pilgrimage

Each year, the Hajj pilgrimage to Mecca, Saudi Arabia attracts about 2–3 million Muslims from different countries around the world in the last month (12th month) of the Arabic calendar (Dhu Al Hijjah) [12]. The risk of infectious diseases among the attendees of Hajj is high. Vaccination against meningococcal disease is mandatory for all, and yellow fever and polio vaccines are also mandatory for pilgrims from endemic countries. Additionally pilgrims are recommended to remain up to date with other vaccines, including seasonal influenza vaccine [17]. However, vaccine uptake among pilgrims is highly variable [18].

We have identified 26 key studies reporting the disease burden at Hajj, of these 15 reported transmission of influenza, occurrence of other infections such as meningococcal and pneumococcal diseases, tuberculosis, and pertussis were reported, and from historical reports, one key publication on cholera has been included.

3.2.1.1. Cholera. Repeated outbreaks of cholera have occurred at Hajj over the centuries [19–21]. Omar listed the outbreaks of cholera occurring between 1877 and 1912; the number of fatalities in those outbreaks ranged from 306 to 32,994 [22]. The largest cholera outbreak happened in 1893, with the fatality rate reaching up to 1000 cases per day [23]. The Hajj assembly in 1893 commenced on 23rd of June, and cholera was declared amongst returned pilgrims in Jeddah on June 29th; by July 13th over 10,000 fatalities were reported from Mecca and Jeddah [24]. Consequently, the Hajj season ended with a total death toll of 32,994 [22]. Several control measures, including quarantine and vaccination, were instituted, and although the disease was ultimately brought under control, the vaccination programme confronted resistance due to cultural, religious and political conflicts [19]. The last known outbreak of cholera occurred in 1989 (Table 2) [12].

3.2.1.2. Meningococcal disease. Two major outbreaks of meningococcal disease are well known: the first was caused by meningococcal serogroup A in 1987 which affected about 2000 pilgrims in various countries, and the next was caused by serogroup W135 in 2000–2001 that affected collectively around 2400 pilgrims across the globe [25]. Extensive molecular investigation revealed that the 1987 outbreak originated in Nepal in 1983–1984, spread to Delhi in 1985 and was brought to Mecca by Hajj pilgrims in 1987 [26], whence it spread to neighbouring countries, and reached Europe, notably UK [27,28]. Within 4 weeks of Hajj, 15 cases were recorded in Qatar, six of them were returned pilgrims and the rest were secondary cases [28]. In the UK alone there were 18 cases of meningococcal A disease among returned pilgrims and 15 additional cases among the contacts [27]. Subsequently, the Saudi Arabian health authority mandated bivalent A and C vaccine for all pilgrims [25].

The 2000–2001 outbreaks have been separately described in a few key reviews [12,25]. Memish and colleagues have provided surveillance data on meningococcal disease (1995–2011)

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Table 2
Vaccine preventable diseases at mass gatherings.

Events	Year	Population (sample size)	Age group (range)	Diseases	No. of cases (%)	No. of fatalities (%)	Comment
Religious MGs							
Hajj [23]	1893	300,000	NR	Cholera	NR	10,000	NR
Kumbh Mela [59]	1945	(300,000)	NR	Cholera	267	157	Voluntary cholera vaccine uptake rate was 0.5%. Pilgrims in whom vaccine uptake was 95%, none had cholera.
	1949–1950	(202,000)	NR	Cholera	39	14	Cholera vaccine uptake was 65%. Of 100,000 pilgrims in whom vaccine uptake was 5%, 202 had cholera, 49 died.
Hajj [27]	1987	10,000	NR	Serogroup A meningococci	18 primary and 15 secondary cases	NR	NR
Hajj [28]	1987	NR	Adults (mean age 40 y) and children (mean age 15 m)	Serogroup A meningococci	6 primary and 9 secondary cases	NR	NR
Hajj [41]	1991	289 305	10–80 y	*S. pneumoniae* Influenza	14 (4.8) 2 (0.7)	NR	NR
	1992	106 456	10–80 y	*S. pneumoniae* Influenza	13 (12.3) 47 (10.3)	NR	
Hajj [52]	1994	(64)	63 (21–91)	*M tuberculosis* *S. pneumoniae*	13 (20.3) 6 (9.4)	NR	The study population are those admitted to hospital with pneumonia
Hajj [29]	2000	60,000	NR	Serogroup W135 meningococci	12 primary cases 78 secondary cases	NR	NR
Hajj [30]	2000	(253) confirmed cases of meningococci	NR	Serogroup W 135 meningococci	93 (37)	NR	NR
Hajj [73]	2000–2001	654 confirmed cases of meningococci in KSA	NR	Meningococcal disease	223 (34) were Hajj pilgrims	NR	NR
Hajj [56]	2002	(358)	48 (16–75)	Pertussis	5 (1.4)	NR	82% (293/358) had immunity against pertussis which can be due to vaccination or prior infection
Hajj [55]	2002	(357)	49.2 (18–75)	*M tuberculosis*	15 of 149 (10) newly seroconverted	NR	BCG vaccine 96.4% (346/357)
Hajj [40]	2003	(115)	NR	Influenza	44 (38)	NR	98% were influenza A; vaccine coverage was 30%.
Hajj [37]	2003	(500)	55% were aged 20–40	Influenza	30 (6)	NR	90% influenza were influenza B; vaccine coverage was 4%.
Hajj [53]	2004	42	Mean age 65	*S. pneumoniae*	12 (29)	NR	NR
Hajj [33]	2005	(483)	51.6 (13–96)	Influenza	47 (9.7)	NR	Influenza A was the dominant strain (77%); vaccine coverage was 30%
Hajj [45]	2005	(202)	44 (1.5–83)	Influenza	28 (14)	NR	Influenza A was the dominant strain (75%); vaccine coverage was 28%
Hajj [54]	2005	(141)	94% aged >50	*S. pneumoniae*	5 (3.5)	NR	NR

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Table 2 (*Continued*)

Events	Year	Population (sample size)	Age group (range)	Diseases	No. of cases (%)	No. of fatalities (%)	Comment
Hajj [46]	2006	UK pilgrims (150)	44 (1.5–83)	Influenza	17 (11)	NR	Influenza A was the predominant type (76%); vaccine uptake was 37%
	2006	Saudi pilgrims (110)	30 (16–85)	Influenza	11 (10)	NR	Influenza A was the dominant type (82%); vaccine uptake was 4%
Hajj [34]	2006–2007	(255)	52 (18–82)	Influenza	25 (9.8)	NR	Influenza A and B circulated equally; vaccine uptake was 85%
WYD 2008 [7]	2008	(227)	21 (12–72)	Influenza	100 (44)	NR	61% were influenza A
WYD 2008 [58]	2008	(242)	26 (12–66)	Influenza	45 of 114 (39%) who were tested had flu	NR	Influenza vaccination rate was 10%
Hajj [44]	2009	(275)	46 (20–70)	Influenza	33 (12)	NR	Influenza B was dominant (61%); vaccine uptake was 100%
Hajj [47]	2009	(305)	49 (24–65)	Influenza A	13 (4.3)	NR	Influenza vaccine 98%
Hajj [43]	2009	Arriving (519)	44.7 (SD 13.9)	Influenza	1 (0.2)	NR	Uptake rates of seasonal and pandemic influenza were 53% and 39% respectively
	2009	Departing (2699)	50 (SD 13.4)	Influenza	9 (0.3)	NR	Influenza A was commoner (78%)
Hajj [36]	2010	(1600)	82% were >40 y	Influenza A	120 (7.5)	NR	Influenza vaccine uptake was 93%
Taizé Meeting [62]	2010	NR	16.5 (9–32)	Measles	13 primary, 24 contacts	NR	Except secondary case who was incompletely vaccinated the rest were unvaccinated
Hajj [48]	2012	(167)	59 (21–83)	Influenza	7 of 70 (10%) who were ill at Hajj had flu	NR	6 of 7 had influenza A, the other had influenza C; Vaccine uptake was 46%.
Hajj [39]	2013	(129)	62 (34–85)	Influenza	10 (7.8)	NR	Predominantly influenza A (90%) circulated; none received seasonal flu vaccine
Hajj [38]	2013	(1038)	35 (18–75)	Influenza	4% (5/112)	NR	All were influenza A; overall flu vaccine uptake was 31%
Hajj [35]	2013	(839)	52 (21–85)	Influenza A	11 (1.3)	NR	Vaccination uptake not recorded
Hajj [42]	2013	(38)	59 (25–83)	Influenza *S. pneumoniae*	6 (15.8) 14 (36.8)	NR	All influenza A
Mass sport gathering							
International Special Olympics Games [63]	1991	40,000–150,000	21 (14–31)	Measles	16 primary and 9 secondary cases	NR	Apart from six cases who had just one dose of vaccination, the rest were not vaccination, although one was age-ineligible.
Winter Olympiad [9]	2002	(188) with upper or lower respiratory symptoms	35 (18–67)	Influenza	36 (19)	NR	Influenza A was the dominant strain (78%); influenza vaccine uptake was 23%

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Table 2 (*Continued*)

Events	Year	Population (sample size)	Age group (range)	Diseases	No. of cases (%)	No. of fatalities (%)	Comment
International Youth Sporting Event [64]	2007	265,000	(10–13)	Measles	7 primary and two generations of secondary cases	NR	While those with no or incomplete vaccination were affected at least two cases received two doses of measles vaccine
Asian Youth Games [8]	2009	66 suspected cases of pandemic flu	NR	Influenza A(H1N1)pdm09	6 (9)	NR	NR
Entertainment events							
'Jam bands' Concert [65]	2003	1200 to 82,000	23 (17–44)	Hepatitis A	25 cases reported in nine US states	NR	Approximately 300 attendees were vaccinated, some were administered post- exposure prophylaxis
Easter Festival [61]	2008	NR	24 (6–69)	Mumps	214	NR	Vaccination status was known for 169 cases, of these 49% were unvaccinated, 40% received one dose and the remaining 11% received two doses
Rock Werchter Festival [66]	2009	69,000	23 (18–45)	Influenza A(H1N1)pdm09	12	NR	NR
Disney Theme Parks [68]	2014	NR	(7 m–70 y)	Measles	42	NR	Of 34 cases who vaccination status was known five received two doses, the rest 29 were unvaccinated, age-ineligible or incompletely vaccinated
International Dog Show [67]	2014	1100	31 (5–52)	Measles	23 primary and 21 likely secondary cases	NR	Excluding five children too young to be vaccinated only 12 of 39 adults were fully vaccinated

NR, not reported; WYD, World Youth Day.

demonstrating that an outbreak affected 223 Hajj pilgrims in Saudi Arabia in 2000–2001 [29,30], Another study, conducted by a separate team of investigators, showed that 37% of meningococcal cases in Mecca, Medina and Jeddah around the Hajj season in 2000 were due to serogroup W135 [30]. The outbreak reached Europe (mostly UK and France) from where some 90 confirmed cases of meningococcal serogroup W135 were reported, of them 12 were in returned pilgrims, 52 in contacts of pilgrims but in the other 26 no definite pilgrim contact was established [29] (Table 2). Many more cases occurred in child contacts of pilgrims than in UK pilgrims themselves. The largest local outbreak in Europe occurred in East London, where of 9 cases of Hajj-associated W135 cases, 8 were children [31].

3.2.1.3. Influenza. The health burden of influenza at Hajj has been reviewed previously [6], but we add several new insights. Historically, influenza among Hajj attendees was first studied in 1975 among Tunisian pilgrims: a serological investigation by complement fixation test revealed an attack rate of influenza of 6.6%. The paper was in French and, therefore, did not meet our inclusion criteria [32]. Subsequently, from 1991 to 2013, the burden of influenza at Hajj was investigated in 16 studies [33–48]. A case series of 7 Austrian pilgrims 5 of whom had influenza immediately after their return from the Hajj in 2014, has been recently published [49]. However, since case reports were not included in our review these data were not included in the summary synthesis. Two separate end points: point prevalence and attack rate (where the denominator is those with respiratory symptoms) have been reported in these studies. The attack rate of influenza ranged from 0.7% among pilgrims with respiratory symptoms in 1991 to 15.8% among pilgrims with bilateral pneumonia in 2013,with a cumulative attack rate of 8.7% [42]. The prevalence ranged from 0.3% in 2009 to 38% in 2003 with a cumulative prevalence of 3.6% [40,43]. The lowest prevalence of influenza (0.4%) was recorded in 2009 (the pandemic year) among arriving and departing international pilgrims at Jeddah airport, however during the next Hajj a much higher prevalence (7.5%) of influenza A was reported by Ashshi and team [36,43].

Both influenza A and B have been reported in these studies. Although most studies demonstrated a predominance of influenza A, circulation of a particular type is not typical of Hajj. For instance, influenza B was the dominant type among international pilgrims during the Hajj 2003 at 90% [37], but among the UK pilgrims influenza A H3N2 (95%) was the dominant strain [40]. While influenza A dominated among international pilgrims during the Hajj 2009 [43], influenza B was most prevalent among pilgrims from Shiraz, Iran [44]. Influenza C was reported only once in a

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French Hajj pilgrim, apparently introduced by a pilgrim from Marseille [48].

Influenza viruses at Hajj were characterised infrequently but where done suggests that circulating strains often do not match with vaccine strains [33,49], however quite often the lineage of influenza B matches with vaccine lineage, although in 2010, the dominant circulating strain was B/Sichuan (24%) which belongs to Yamagata lineage while the vaccine strain was a B/Brisbane/60/2008-like virus of Victoria lineage [50].

3.2.1.4. Pneumococcal disease. Pneumonia is the main cause of hospitalisation during Hajj [51]. *Streptococcus pneumoniae* was isolated from 4.8% and 12.3% of symptomatic pilgrims in the Hajj 1991 and 1992 respectively [41]. Among pilgrims admitted to Mecca hospitals with pneumonia during the Hajj 1994, *S. pneumoniae* was detected in 9.4% cases [52]. *S. pneumoniae* accounted for 19% (12/64) of bacteriological isolates from patients admitted to ICU with sepsis during the Hajj 2004 [53]. In 2005, out of 141 cases that were admitted to three tertiary hospitals in Mecca with clinically suspected pneumonia, a bacterial pathogen was detected in 76 (54%) patients, and *S. pneumoniae* was responsible for 3.5% (5/141) of likely pneumonia cases [54]. In 2013, among 38 patients who were hospitalised with bilateral pneumonia a microbiological aetiology was established in 26 (68.4%) cases, and *S. pneumoniae* was identified in 14 (53.8%) positive samples (Table 2) [42].

3.2.1.5. Tuberculosis. Two studies reported the burden of tuberculosis among Hajj pilgrims. In a cross sectional study conducted among 64 patients who were admitted to two tertiary hospitals in Mecca during the Hajj 1994, *Mycobacterium tuberculosis* accounted for 20.3% [52]. A separate study by Wilder-Smith and colleagues found that 10% of returned Singaporean pilgrims had sero-conversion for *M tuberculosis* (Table 2) [55].

3.2.1.6. Other infections. Smallpox was common among pilgrims in the 1940s [19,22]. The last outbreak occurred in 1949 with 545 cases and 198 fatalities reported [22]. In a cohort of 358 Singaporean pilgrims during the Hajj 2002, 1.4% were found to acquire pertussis [56]. In addition, the risk of other VPDs, including the likely risk of hepatitis A and B have been discussed elsewhere (Table 2) [20,57].

3.2.2. World Youth Day

WYD is a Catholic youth festival which is considered to be the largest MG of young people of Catholic faith in the world. It is held every three years in different countries, attracting up to 5 million pilgrims. The last WYD was held in Brazil in 2013 with about 4 million attendees, and the next event will take place in July 2016 in Kraków, Poland. We found two studies reporting outbreaks of influenza infection at WYD 2008 in Australia [7,58]. Van Hal et al. reported that among 114 ill pilgrims who underwent diagnosis for influenza at a single clinic at WYD 2008, Sydney, 39.5% had influenza [58]. At the same event, Blyth et al. described the epidemiological and virological features of an influenza outbreak among pilgrims. The attack rates of influenza A and B were 27% (61/227) and 17% (39/227) respectively among pilgrims who sought medical treatment for influenza-like illness (ILI) (Table 2) [7].

3.2.3. Kumbh Mela

Kumbh Mela is a Hindu religious festival held in India every 3 years. It attracts around 100 million pilgrims and therefore is considered to be the largest MG in the world. Epidemics of water and food borne illnesses and stampede injuries have been reported [10]. We include one key report on cholera at Kumbh Mela [59].

A massive cholera outbreak from Kumbh Mela that took place in Haridwar in 1867 reached Peshawar, Kabul, Persia, Turkey, Ukraine and other parts of Russia killing almost one million people [60]. Cholera vaccine was made compulsory in 1945. In a sample of 250,000 pilgrims, Rogers et al. demonstrated that no outbreak occurred when 75% of the participants were vaccinated. A lower vaccination rate of 65% was associated with not fully containing the disease although allegedly, a considerable reduction in morbidity and mortality occurred in a sample of >2,000,000 pilgrims. An extremely low vaccination uptake (0.5% or 5%) hardly had any beneficial impact (Table 2) [59].

3.2.4. Easter Festival

An outbreak of mumps has been reported among young adults after an Easter Festival in Austria in 2006 with 214 cases, which was attributed to incomplete protection from just a single dose of MMR vaccine [61].

3.2.5. Taizé Festival

Taizé meeting is a catholic festival held every year in France. In 2010 a measles outbreak was reported among returned travellers from Germany. A total of 13 primary cases, and 17 secondary cases and seven tertiary cases were identified. Apart from a secondary case who received only one dose of measles-containing vaccine all others were unvaccinated [62].

3.3. Sport events

3.3.1. International Special Olympic Games

The Special Olympic Games is an international sporting event for athletes with disabilities and attracts more than 40,000 attendees. It occurs every two years and alternates between summer and winter games. The last event took place in 2013 in Pyeongchang, South Korea, and the next one is in 2015 in Los Angeles, USA. We identified one relevant study, which recorded a measles outbreak in 1991 [63].

The Special Olympic World Games 1991 was held in Minneapolis, USA where measles was introduced by a track and field athlete from Argentina. Subsequently 16 cases of measles occurred in athletes, spectators, and volunteers about 2 weeks after the event. An additional nine cases resulted from subsequent transmission. All cases were aged ≤30 years. Eighteen of 24 cases were not up to date with measles vaccine (Table 2) [63].

3.3.2. Winter Olympiad

The Winter Olympic Games is a large international sport event, which is held every four years in different countries; the Winter Paralympic Games follows a few weeks later essentially at the same venues. As it occurs in winter, the Games may coincide with seasonal epidemics of influenza. We found only one study reporting an influenza outbreak during Winter Olympic and Paralympic Games of 2002 [9].

The Winter Olympiad 2002 took place in Salt Lake City, Utah, USA. A prospective surveillance for influenza was conducted among athletes, staff and 'Olympic family' members. Of 188 patients meeting the criteria for potential influenza 36 (19%) had influenza established by viral culture; influenza A was the dominant strain. Oseltamivir was administered to all patients with ILI or confirmed influenza and their close contacts (Table 2) [9].

3.3.3. Asian Youth Games

Asian Youth Games is an Asian international sport event which is held four yearly. The first Asian Youth Games took place in Singapore in 2009 in the midst of pandemic influenza, where around 1200 athletes participated from 43 countries. Singaporean medical experts published their experience of managing the event [8]; which took place from 29 June to 7 July, 2009. By the opening day of the games, Singapore had already reported 600 cases of H1N1

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pandemic strain of 2009. During the event, out of 66 suspected cases, 6 (9%) were confirmed to have influenza A(H1N1)pdm09 [8]. The investigators opined that an effective contingency plan, nimble risk assessment, excellent communication and partnership helped contain the outbreak (Table 2).

3.3.4. International Youth Sport Event

An outbreak of measles occurred among attendees of an international youth sport event in Pennsylvania in August 17–26, 2007. About 265,000 participants from Americas, Europe and Asia attended this event. The outbreak of measles comprised seven primary and two generations of secondary cases. While those with incomplete vaccination or no record of vaccination were affected at least two cases had records of obtaining two doses of measles vaccine [64].

3.4. Entertainment MGs

3.4.1. Music concert 'Jam bands'

'Jam bands' is an outdoor music festival that involves camping on established or unplanned campgrounds where the sanitary condition is below optimum standard. The bands performing at these concerts and festivals attract between 1200 and 82,000 fans that travel from one concert to another, sometimes over several months often living on foods from unofficial vendors. Concerts may go on for days. An outbreak of hepatitis A occurred among participants of "Jam bands" music festivals in the USA in 2003. A total of 25 cases with a median age of 23 (range 17–44 years) were identified among the participants from nine US states [65]. Following initial investigation, vaccination and education campaigns were held in an attempt to curb the outbreak.

3.4.2. Rock Werchter Festival

Rock Werchter is a Belgian annual music festival held in the village of Werchter, near Leuven. It is considered as one of the four biggest annual rock music festivals in Europe. It attracts around 80,000 guests over a period of four days mostly in the beginning of July or at the end of June. In 2009, Belgium hosted the rock festival in the first week of July with about 69,000 attendees at the peak of pandemic influenza. Gutiérrez et al. recorded 12 confirmed cases of influenza A(H1N1)pdm09 out of 30 (40%) young individuals with ILI symptoms (Table 2) [66]. Two additional cases were reported from the Netherlands and Luxembourg making a total of 14 cases linked to the event. The index case was a visitor from Israel.

3.4.3. International Dog Show

Every year, the International Dog Show attracts thousands of participants from different countries. It is one of the largest dog shows in the world, and takes place in different locations every year. We found one study reporting a measles outbreak linked to a recent dog show festival that took place on 8 and 9 November, 2014 in Vrtojba, Slovenia [67]. About 1100 participants were said to have attended the show. A total of 44 cases linked to the dog show were recorded; of these 23 were infected at the dog show, while the rest were presumably secondary or tertiary cases. At least five unvaccinated children were affected (Table 2) [67].

3.4.4. Disney Theme Parks

Disney Parks are among the biggest theme parks in the world, attracting around 16 million guests of all ages from different countries round the year. Recently a measles outbreak had been recorded in Disney Parks [68]. At the end of December 2014, US public health authority reported 42 cases of measles linked to the Disneyland Park, and Disney California Adventure Park. Individuals of all age groups, including five employees, were affected. Of 34 individuals whose vaccination status was known, 28 (82%) were unvaccinated or too young to be vaccinated (Table 2) [68].

4. Discussion

The review canvases the many infectious risks associated with MGs and demonstrates that a variety of VPDs occur at these large public events. The Hajj pilgrimage is the most studied MG, and currently respiratory tract infections are the commonest health hazards among the attendees of most MGs.

Influenza, meningococcal disease and measles are the most commonly identified infectious diseases. This is likely because the uniquely high 'contact rates' in a crowded milieu allow the airborne microbes to transmit easily [69], and their relatively shorter incubation period allows their earlier detection.

Influenza has been reported from both sport (Winter Olympics, Asian Youth Games) and religious (Hajj, WYD) MGs. Although travellers (e.g., Hajj pilgrims) are encouraged to obtain influenza vaccine before they travel to the event, the uptake is variable, but encouragingly increasing in recent years [70]. Consequently, the rate of ILI is decreasing, and a meta-analysis of observational studies demonstrated a significant effect of vaccination on laboratory-proven influenza although the overall effect of the vaccine on disease is yet to be seen and controlled studies exploring the efficacy of influenza vaccine among attendees of MGs are not available [18,71].

Although outbreaks of meningococcal disease have been considered typical of Hajj [25]; they have occurred elsewhere e.g., an outbreak at the 2004 European Football Tournament (EURO 2004) [72]. Mandatory meningococcal vaccination of pilgrims and residents living around pilgrimage sites reduced the incidence of the invasive disease not only among travellers but also among the native population [73]; however the polysaccharide vaccine, the most widely used vaccine [74], is unlikely to impact pharyngeal carriage of meningococci and thus would not prevent transmission of pathogenic strains to pilgrims' contacts [75]. A conjugate quadrivalent vaccine is more likely to prevent acquisition and perhaps also clears throat carriage and thus provide community protection [76]. In view of the changing epidemiology of the disease in countries that send pilgrims to Hajj, some experts have stressed the importance of meningococcal B vaccine for Hajj pilgrims [77].

Outbreaks of measles linked to both sport and entertainment events have been reported [63,67,68]. In regions with interrupted endemic measles transmission, imported and import-related cases account for virtually all measles in the area, but in regions with high vaccine uptake (≥95%) the transmission of the disease was very limited [78]. Consequently, almost all measles outbreaks at MGs have occurred among unvaccinated or incompletely vaccinated individuals. We are not aware of any outbreak of measles in association with Hajj pilgrimage although an outbreak among attendees of a catholic festival in Taizé, France is reported [62]. Relatively older age of Hajj pilgrims and a high level of immunity among the attendees could be the possible explanation. Outbreak of mumps resulting from incomplete protection following a single dose of MMR vaccine has been reported at MG setting [61]. Similarly, an outbreak of rubella occurred among unvaccinated young male cadets deployed at Hajj in 1995, but pilgrims were not affected [79].

Large cholera outbreaks are no more reported at MGs [20]; however other water and foodborne diseases preventable by vaccine are occasionally reported. Occurrence of hepatitis A following 'Jam bands' concerts in the USA is one such example. Rafiq and colleagues discussed that although hepatitis A has not yet been reported from Hajj, many pilgrims engage in high risk behaviours such as buying foods from unregulated vendors, communal use of

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toilets, avoidance of scented soap and drinking iced water [65]. Similarly, the risk of food and waterborne diseases, both vaccine-preventable, and non-preventable infections, is high. For instance, recent qualitative observation and epidemiological surveillance at Kumbh Mela suggest that the level of water sanitation and hygiene of bathing water of Kumbh fair is substandard [80]. Similarly, food borne outbreaks have been reported at other mass events such as FIFA World Cup 2006 in Germany, where an outbreak of norovirus affected over 60 attendees [81].

VPDs that transmit through the sexual or parenteral route such as hepatitis B and human papilloma virus are not widely reported. However there is a considerable risk of spread of these infections at MGs. For instance, the risk of transmission of hepatitis B among athletes of contact sports can be high [82,83]. Similarly, pilgrims at Hajj and Kumbh Mela engage in other activities (e.g., communal use of razors for head shaving) that may increase the risk of parenteral hepatitis [57]. Due to the prolonged incubation period of these infections, systematic case finding is challenging.

Transmission of drug resistant pathogens such as *S. pneumoniae* is a growing risk at MGs [84]. It is postulated that about one third of pneumococcal isolates are resistant to penicillin and one tenth resistant to third generation cephalosporins [51,85]. Circulation of highly penicillin resistant *S. pneumoniae* in Spain at the time of Barcelona Olympics in 1992 was of concern [86]. Conjugate pneumococcal vaccines can play a role in preventing transmission of antibiotic resistant pneumococci [87].

The type of MGs and the activities of the participants can influence the epidemiology of VPDs at MGs. For instance, respiratory infections including pneumonia (such as influenza or pneumococcal pneumonia) are the common health hazards among Hajj pilgrims, half of whom are older than 50 years of age [88,89]. On the other hand, measles and mumps are common among attendees of sport and fun events which are primarily attended by the young [18,51]. Influenza, on the other hand, is common in both religious and sport events [6]. Local epidemiology of the host country also influences the disease pattern. For instance, outbreaks of cholera at Kumbh Mela festivals in the Ganges basin are unsurprising since historically the Ganges is known to be a source of explosive outbreaks of cholera [10,90]. Similarly, the pharyngeal carriage rate of meningococci in the most crowded part of Mecca was as high as 86% during the interim period between the two major outbreaks of meningococcal disease at Hajj [91].

From the examples of intercontinental outbreaks of meningococcal disease, and globalisation of respiratory pathogens from Hajj [2,26], outbreaks of cholera from Kumbh Mela and Hajj [60], and international spread of measles from MGs [78], it is apparent that spread of infectious disease can have far reaching ramifications. Vaccination can play a key role in the prevention of VPDs. Pre-departure vaccination of a cohort of Egyptian pilgrims brought the prevalence of pandemic influenza A(H1N1)pdm09 to nil during the Hajj in October 2009 [92]. Furthermore, good planning and effective surveillance systems are other key components of controlling infectious disease at MGs. For example, by planning well and strengthening public health services the London 2012 Olympics proceeded without major incident, similarly Hajj and Asian Youth Games were organised during 2009 pandemic influenza with no discernable upsurge of the disease [8,93].

In recent years, experts have recommended vaccines for travellers who intended to attend mass gathering events. For instance,

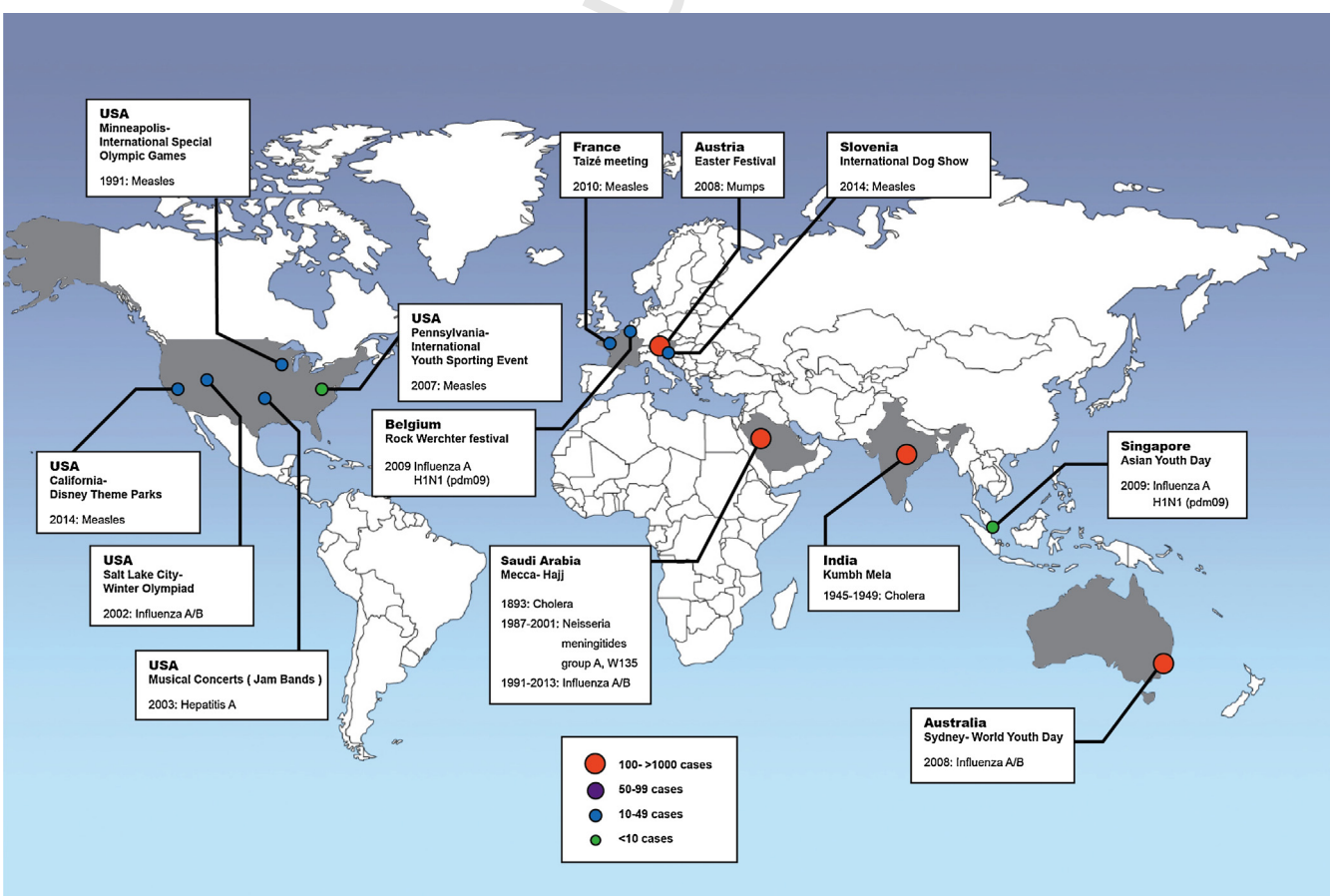

Fig. 2. Map showing the key location and features of the mass gatherings.

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for attendees of the Olympic and Paralympic Games in Beijing in 2008 several vaccines, including hepatitis A and B, influenza and typhoid were recommended [94]. In the same year, the attendees of European Football Tournament 2008 (EURO 2008) that was held in Switzerland, were advised to be immunised against measles and tick-borne encephalitis [95]. Attendees of the FIFA World Cup 2010, that took place in South Africa, were recommended to receive vaccinations against hepatitis A and B, influenza, typhoid, and MMR [96]. Moreover, because of the higher risk of meningococcal disease in Africa, some experts additionally recommended the quadrivalent meningococcal vaccine for travellers to South Africa [97]. Travellers to the FIFA 2014, Brazil, were advised to remain up-to-date with routine vaccines, including hepatitis A, influenza, MMR, tetanus, diphtheria, pertussis, pneumococcal and varicella [98]. Further recommendation has also been issued for travellers to Brazil to participate in the 2016 Olympic Games; the spectators have been particularly advised to receive influenza and measles vaccines [99].

Despite the recognised benefits of vaccination, uptake among attendees of MGs is variable and often suboptimal. For instance, Memish et al. demonstrated that despite compulsory requirement, about 7% pilgrims did not receive appropriate meningococcal vaccine [74], and concerns were raised about the issuing of false meningococcal certificates to some South African pilgrims a decade ago [100]. Studies found that lack of knowledge about vaccine was an important barrier for vaccination [43,101]. Education of health professionals and involvement of tour group leaders might improve vaccination coverage [102].

To our knowledge, this is the first review that has assessed the burden of VPDs at MGs. VPDs can occur at diverse types of events including religious, sports and entertainment events. An important limitation is that studies reporting syndromic infections were excluded from our review and most studies reported only syndromic data. Moreover, in this review we did not evaluate the effectiveness of vaccines. The limited available data suggest that the vaccines are effective against meningococcal disease, influenza and cholera [18,19,59,71,73]. Studies are needed to quantify the effects of vaccines more precisely.

5. Conclusion

VPDs have been reported at many MGs. Infections not only affect the participants of MGs but also their contacts. Despite recommendations issued by experts, event organisers or international public health bodies the uptake of vaccines is variable and generally low, and there is a dearth of data on the effectiveness of vaccines at MGs. Research should be inspired to address those gaps.

Conflict of interest

Professor Robert Booy has received funding from Baxter, CSL, GSK, Merck, Novartis, Pfizer, Roche, Romark and Sanofi Pasteur for the conduct of sponsored research, travel to present at conferences or consultancy work; all funding received is directed to research accounts at The Children's Hospital at Westmead. Dr Harunor Rashid received fees from Pfizer and Novartis for consulting or serving on an advisory board. The other authors have no competing interests to declare.

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